

See Singapore's heat now. Plan for tomorrow.

Singapore at its best, with care at its heart.

01

PROBLEM

Urban heat in dense residential estates disproportionately threatens older residents, but live exposure, vulnerability and cooling access are rarely viewed together.

02

SOLUTION

One interactive map combines live station heat, an ML-based 24-hour forecast, neighbourhood risk tiers and cooling what-if scenarios for targeted alerts.

03

DATASETS

NEA temperature, humidity and rainfall; SingStat age structure; URA / HDB geography; MOH eldercare; NParks / LTA green networks.

04

DATABRICKS ARCHITECTURE

NEA APIs and reference files land in Delta Lake; notebooks build hourly features; MLflow tracks forecast and risk models; Jobs refresh Gold outputs for AI/BI.

COMMUNITY HEAT RISK MAP

LIVE HEAT INDEX

Current heat index across the station network.

AREA RISK SCORE

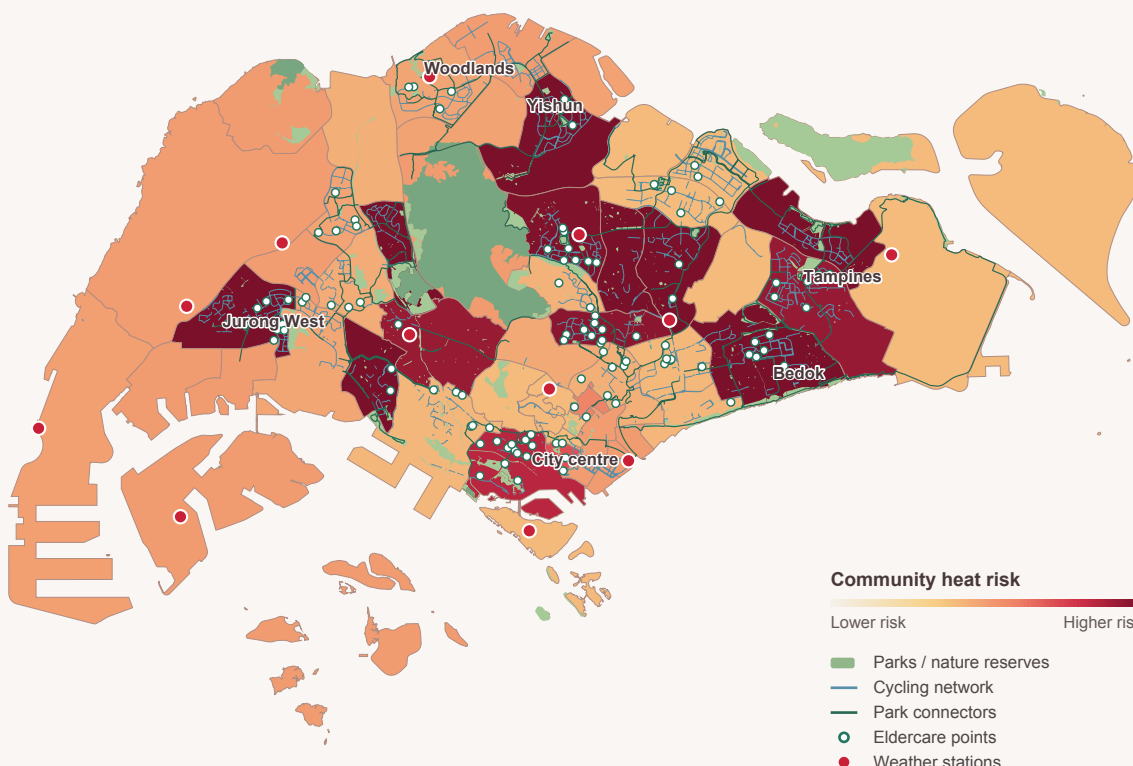
Heat + elderly density + access to cooling.

COOLING SIMULATION

Compare added cooling points and corridor shade.

24H HEAT FORECAST

V4 hourly heat-risk outlook for the next 24 hours.



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PRE-RELEASE WEBSITE

promelin.github.io/heatguard-sg